

Extra.

IIIT, Bhubaneswar
Fundamentals of Image Processing

(7th Sem. ETC)

Mid Semester Examination

Duration: 90 mins.

Full Marks: 30

(Answer any six questions including Q1. All questions carry equal marks)

[1+2+2=5]

1. (a) What is alpha trimmed mean filter?
(b) Explain power law transformation with its applications.
(c) Determine a 5×5 Gaussian kernel using PASCAL's triangle method.
2. A 4×4 , 4-bits/pixel image $f(m,n)$ is passed through point wise intensity transformation $g(m,n) = \text{round}(10 \times \sqrt{f(m,n)})$. Determine the output image $g(m,n)$ when

$$f(m,n) = \begin{bmatrix} 12 & 8 & 4 & 9 \\ 10 & 5 & 3 & 6 \\ 8 & 12 & 9 & 13 \\ 4 & 12 & 9 & 10 \end{bmatrix}$$

3. (a) Explain m-adjacency and its advantage with an example.

[3+2=5]

- (b) Consider the image segment
(p) $\begin{matrix} & 3 & 1 & 2 & 1 \\ & 2 & 2 & 0 & 2 \\ & 1 & 2 & 1 & 1 \\ (p) & 1 & 0 & 1 & 2 \end{matrix}$ (q)
of the shortest 4-, 8-, and m-path between p and q. . Let $V = \{1,2\}$ and compute the length

4. Describe the first derivative and second derivative image sharpening filters with its advantages and disadvantages.
5. Prove that the mean filter used in image processing is a low pass filter.
6. What condition needs to be satisfied by distance measures? Explain various distance measure between two pixels.
7. Explain the order statistic filters with its applications.